1. Two Sum

Problem Statement

Given an array of integers `nums` and an integer `target`, return *indices of the two numbers such that they add up to target*.

You may assume that each input would have **exactly one solution**, and you may not use the *same* element twice.

You can return the answer in any order.

Examples

> ### Example 1
>
> **Input:** `nums = [2,7,11,15], target = 9`
>
> **Output:** `[0,1]`
>
> **Explanation:** Because `nums[0] + nums[1] == 9`, we return `[0, 1]`.

> ### Example 2
>
> **Input:** `nums = [3,2,4], target = 6`
>
> **Output:** `[1,2]`

Example 3

Input: `nums = [3,3], target = 6`

Output: `[0,1]`

Constraints

- `2 <= nums.length <= 10^4`
- `-10^9 <= nums[i] <= 10^9`
- `-10^9 <= target <= 10^9`
- **Only one valid answer exists.**

Follow-up: Can you come up with an algorithm that is less than $O(n^2)$ time complexity?